

Exersien 4

1)

Vi finner posisjonen til en satellitt med tid t_0

$$\vec{r} = [-3126.903, -4994.661] \text{ km}$$

$$h = |\vec{r}| - r_p = 3589.126706... \text{ km}$$

(satellitt har samme høyde)

2)

Vi har:

$$\frac{\text{m}^3}{\text{kg s}^2}$$

$$F_G = m \frac{v^2}{r}$$

$$G \frac{Mm}{r^2} = m \frac{v^2}{r}$$

$$v = \sqrt{\frac{GM}{r}}$$

Vi setter inn ω for:

$$v = 2.8243... \frac{\text{km}}{\text{s}}$$

Exersien 4)

$$t_0 = 209.83096915$$

$$4.3 \quad \vec{r}_1 = [-3339.883, -4854.827] + 209.80888235 \text{ km}$$

$$\vec{r}_2 = [-465.016, -5874.345] + 209.80569035 \text{ km}$$

$$\phi_i = \Delta t_i \cdot \omega = (t_0 - t_i) \cdot \omega$$

$$\phi_1 \pm \phi_2 = \theta_+$$

$$\phi_1 = \arctan\left(\frac{S_y}{S_x}\right) + \pi$$

Vi vet det er $\pm \pi$ fordi satellitt er i tredje kvadrant, det sin $S_x, S_y < 0$ så arctan kommer i første kvadrant

for ϕ_2 finner vi cos setningen

$R^2 = R_1^2 + R_2^2 - 2R_1R_2\cos(\alpha)$ med avstanden mellom satellitt og ω og ω og sentrum satellitt og radius planet

er vår avstand

$$\phi_2 = \arccos\left(\frac{-R_1^2 - R_2^2 + R_p^2}{2R_1R_2}\right)$$

Regner ut i python for hvert ship i for

$$\begin{array}{c|c} \theta_+ & 5.748602 \\ \theta_- & 2.470989 \end{array} \quad \begin{array}{c} S_1 \\ S_2 \end{array} \quad \begin{array}{c} 6.795292 \\ 2.420997 \end{array}$$

ser at eneste løsning som funker for begge er 2.4709. radianer men de har litt variasjon.

finner $x = r_p \cos(\theta)$, $y = r_p \sin(\theta)$

$$(x, y) = (-1805.537, 1932.187)$$

4.4) (system er satellitt)

Grise (O)

sent (A)

$t=0$ $t'=0$

$t=?$ $t'=t_s$

motat (B)

$t=t_p$ $t'=0$

Tenker 2 stell, et på overflaten med $r=r_p$ og ett på høyde med satellittene $r=r_s$ sammenligner begge med en langt vekke observator

$$\Delta S_{\text{langt vekk}}^2 = \Delta S_{\text{satellitt}}^2$$

$$\left(1 - \frac{2m}{r}\right) \Delta t_{\text{langt}}^2 - r^2 \Delta \phi^2 = \Delta t_{\text{stell}}^2$$

hvor jeg har brukt at satellitt aldri forandrer høyde (sirkelbane)

$$\left(1 - \frac{2m}{r_{\text{stell}}}\right) - \left(\frac{\Delta \phi_{\text{stell}}}{\Delta t_{\text{langt}}}\right)^2 = \left(\frac{\Delta t'_{\text{stell}}}{\Delta t_{\text{langt}}}\right)^2$$

gjør dette med klokken på bakken og satellitt

hvor Δt_{langt} er samme Δt for begge

$$1) \left(1 - \frac{2m}{r_s}\right) - \left(\frac{\Delta \phi_{\text{stell}}}{\Delta t_{\text{langt}}}\right)^2 = \left(\frac{\Delta t'_s}{\Delta t_{\text{langt}}}\right)^2$$

$$2) \left(1 - \frac{2m}{r_p}\right) - \left(\frac{\Delta \phi_p}{\Delta t_{\text{langt}}}\right)^2 = \left(\frac{\Delta t_p}{\Delta t_{\text{langt}}}\right)^2$$

$$2) \Rightarrow \Delta t_{\text{langt}}^2 = \frac{\Delta t_p^2}{1 - \frac{2m}{r_p}}$$

$$\text{Setter inn i 1)}$$

$$\left(1 - \frac{2m}{r_s}\right) - v_{\theta}^2 = 1 - \frac{2m}{r_p} \left(\frac{\Delta t'_s}{\Delta t_p}\right)^2$$

$$\Delta t_s^2 = \left(\frac{1 - \frac{2m}{r_s} - v_{\theta}^2}{1 - \frac{2m}{r_p}}\right) \Delta t_p^2$$

$$\Delta t_s = \sqrt{\frac{1 - \frac{2m}{r_s} - v_{\theta}^2}{1 - \frac{2m}{r_p}}} \Delta t_p$$

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